Name: **Jash Chauhan**

Section: **D – (D1)**

EXP: 2

Aim: **Consider the university database system designed and created in practical 1. Write and execute SQL queries for performing the following.**

i. Create various tables as per the schema diagram of the practical 1.

```
=> CREATE TABLE UNIVERSITY (
   university_id INT PRIMARY KEY,
   name VARCHAR(100) NOT NULL
);

CREATE TABLE DEPARTMENT (
   dept_id INT PRIMARY KEY,
   name VARCHAR(50) NOT NULL,
   building VARCHAR(50),
   budget DECIMAL(12, 2),
   university_id INT,
   FOREIGN KEY (university_id) REFERENCES UNIVERSITY(university_id)
);

CREATE TABLE INSTRUCTOR (
   instructor_id NUMBER PRIMARY KEY,
   name VARCHAR2(50) NOT NULL,
   dept_id NUMBER,
   is_hod CHAR(1) DEFAULT 'N' CHECK (is_hod IN ('Y', 'N')),
   salary NUMBER(10, 2),
   CONSTRAINT fk_instructor_dept
      FOREIGN KEY (dept_id)
```

```
    REFERENCES DEPARTMENT(dept_id)
);

CREATE TABLE COURSE (
  course_id VARCHAR(8) PRIMARY KEY,
  title VARCHAR(100) NOT NULL,
  dept_id INT,
  credits INT,
  FOREIGN KEY (dept_id) REFERENCES DEPARTMENT(dept_id)
);

CREATE TABLE SECTION (
  section_id INT PRIMARY KEY,
  course_id VARCHAR(8),
  sec_id INT,
  semester VARCHAR(6),
  year INT,
  building VARCHAR(50),
  room_number VARCHAR(10),
  time_slot_id VARCHAR(4),
  FOREIGN KEY (course_id) REFERENCES COURSE(course_id)
);

CREATE TABLE TEACHES (
  instructor_id INT,
  section_id INT,
  PRIMARY KEY (instructor_id, section_id),
  FOREIGN KEY (instructor_id) REFERENCES INSTRUCTOR(instructor_id),
  FOREIGN KEY (section_id) REFERENCES SECTION(section_id)
);
```

```
CREATE TABLE STUDENT (
   student_id INT PRIMARY KEY,
   name VARCHAR(50) NOT NULL,
   tot_cred INT
);

CREATE TABLE TAKES (
   student_id INT,
   section_id INT,
   grade VARCHAR(2),
   PRIMARY KEY (student_id, section_id),
   FOREIGN KEY (student_id) REFERENCES STUDENT(student_id),
   FOREIGN KEY (section_id) REFERENCES SECTION(section_id)
);

CREATE TABLE TIME_SLOT (
   time_slot_id VARCHAR2(4) PRIMARY KEY,
   day CHAR(1),
   start_time DATE,
   end_time DATE
);

CREATE TABLE CLASSROOM (
   building VARCHAR(50),
   room_number VARCHAR(10),
   capacity INT,
   PRIMARY KEY (building, room_number)
);
```

ii. Insert minimum 7 rows in each table.

```
=> INSERT INTO UNIVERSITY (university_id, name) VALUES  (1, 'Example University');
```

```
INSERT INTO UNIVERSITY (university_id, name) VALUES  (2, 'Another University');
INSERT INTO UNIVERSITY (university_id, name) VALUES  (3, 'Third University');
INSERT INTO UNIVERSITY (university_id, name) VALUES  (4, 'Fourth University');
INSERT INTO UNIVERSITY (university_id, name) VALUES  (5, 'Fifth University');
INSERT INTO UNIVERSITY (university_id, name) VALUES (6, 'Sixth University');
INSERT INTO UNIVERSITY (university_id, name) VALUES  (7, 'Seventh University');

INSERT INTO DEPARTMENT (dept_id, name, building, budget, university_id) VALUES
(1, 'Computer Science', 'Taylor', 100000, 1);
INSERT INTO DEPARTMENT (dept_id, name, building, budget, university_id) VALUES
(2, 'Mathematics', 'Watson', 90000, 1);
INSERT INTO DEPARTMENT (dept_id, name, building, budget, university_id) VALUES
(3, 'Physics', 'Newton', 95000, 1);
INSERT INTO DEPARTMENT (dept_id, name, building, budget, university_id) VALUES
(4, 'Biology', 'Darwin', 85000, 1);
INSERT INTO DEPARTMENT (dept_id, name, building, budget, university_id) VALUES
(5, 'Chemistry', 'Curie', 92000, 1);
INSERT INTO DEPARTMENT (dept_id, name, building, budget, university_id) VALUES
(6, 'English', 'Shakespeare', 80000, 1);
INSERT INTO DEPARTMENT (dept_id, name, building, budget, university_id) VALUES
(7, 'History', 'Roosevelt', 78000, 1);

INSERT INTO INSTRUCTOR (instructor_id, name, dept_id, is_hod, salary)
VALUES (1, 'John Doe', 1, 'Y', 80000);
INSERT INTO INSTRUCTOR (instructor_id, name, dept_id, is_hod, salary)
VALUES (2, 'Jane Smith', 1, 'N', 75000);
INSERT INTO INSTRUCTOR (instructor_id, name, dept_id, is_hod, salary)
VALUES (3, 'Bob Johnson', 2, 'Y', 82000);
INSERT INTO INSTRUCTOR (instructor_id, name, dept_id, is_hod, salary)
VALUES (4, 'Alice Brown', 2, 'N', 76000);
INSERT INTO INSTRUCTOR (instructor_id, name, dept_id, is_hod, salary)
```

```
VALUES (5, 'Charlie Davis', 3, 'Y', 81000);

INSERT INTO INSTRUCTOR (instructor_id, name, dept_id, is_hod, salary)

VALUES (6, 'Eva Wilson', 3, 'N', 77000);

INSERT INTO INSTRUCTOR (instructor_id, name, dept_id, is_hod, salary)

VALUES (7, 'Frank Miller', 4, 'Y', 79000);


INSERT INTO COURSE (course_id, title, dept_id, credits) VALUES

('CS-101', 'Introduction to Programming', 1, 3);

INSERT INTO COURSE (course_id, title, dept_id, credits) VALUES

('CS-201', 'Data Structures', 1, 4);

INSERT INTO COURSE (course_id, title, dept_id, credits) VALUES

('MATH-101', 'Calculus I', 2, 4);

INSERT INTO COURSE (course_id, title, dept_id, credits) VALUES

('MATH-201', 'Linear Algebra', 2, 3);

INSERT INTO COURSE (course_id, title, dept_id, credits) VALUES

('PHY-101', 'Mechanics', 3, 4);

INSERT INTO COURSE (course_id, title, dept_id, credits) VALUES

('PHY-201', 'Quantum Physics', 3, 4);

INSERT INTO COURSE (course_id, title, dept_id, credits) VALUES

('BIO-101', 'Introduction to Biology', 4, 4);


INSERT INTO SECTION (section_id, course_id, sec_id, semester, year, building, room_number,
time_slot_id) VALUES  (1, 'CS-101', 1, 'Fall', 2023, 'Taylor', '3128', 'A');

INSERT INTO SECTION (section_id, course_id, sec_id, semester, year, building, room_number,
time_slot_id) VALUES (2, 'CS-201', 1, 'Spring', 2024, 'Taylor', '3114', 'B');

INSERT INTO SECTION (section_id, course_id, sec_id, semester, year, building, room_number,
time_slot_id) VALUES (3, 'MATH-101', 1, 'Fall', 2023, 'Watson', '2230', 'C');

INSERT INTO SECTION (section_id, course_id, sec_id, semester, year, building, room_number,
time_slot_id) VALUES (4, 'MATH-201', 1, 'Spring', 2024, 'Watson', '2245', 'D');

INSERT INTO SECTION (section_id, course_id, sec_id, semester, year, building, room_number,
time_slot_id) VALUES (5, 'PHY-101', 1, 'Fall', 2023, 'Newton', '1320', 'E');

INSERT INTO SECTION (section_id, course_id, sec_id, semester, year, building, room_number,
time_slot_id) VALUES (6, 'PHY-201', 1, 'Spring', 2024, 'Newton', '1310', 'F');
```

```
INSERT INTO SECTION (section_id, course_id, sec_id, semester, year, building, room_number,
time_slot_id) VALUES (7, 'BIO-101', 1, 'Fall', 2023, 'Darwin', '2110', 'G');

INSERT INTO TEACHES (instructor_id, section_id) VALUES (1, 1);
INSERT INTO TEACHES (instructor_id, section_id) VALUES (2, 2);
INSERT INTO TEACHES (instructor_id, section_id) VALUES (3, 3);
INSERT INTO TEACHES (instructor_id, section_id) VALUES (4, 4);
INSERT INTO TEACHES (instructor_id, section_id) VALUES (5, 5);
INSERT INTO TEACHES (instructor_id, section_id) VALUES (6, 6);
INSERT INTO TEACHES (instructor_id, section_id) VALUES (7, 7);

INSERT INTO STUDENT (student_id, name, tot_cred) VALUES (1, 'Mike Johnson', 60);
INSERT INTO STUDENT (student_id, name, tot_cred) VALUES (2, 'Sarah Lee', 45);
INSERT INTO STUDENT (student_id, name, tot_cred) VALUES (3, 'Tom Brown', 30);
INSERT INTO STUDENT (student_id, name, tot_cred) VALUES (4, 'Emily Davis', 75);
INSERT INTO STUDENT (student_id, name, tot_cred) VALUES (5, 'David Wilson', 90);
INSERT INTO STUDENT (student_id, name, tot_cred) VALUES (6, 'Lisa Anderson', 15);
INSERT INTO STUDENT (student_id, name, tot_cred) VALUES (7, 'Chris Taylor', 105);

INSERT INTO TAKES (student_id, section_id, grade) VALUES (1, 1, 'A');
INSERT INTO TAKES (student_id, section_id, grade) VALUES  (1, 3, 'B');
INSERT INTO TAKES (student_id, section_id, grade) VALUES (2, 2, 'A');
INSERT INTO TAKES (student_id, section_id, grade) VALUES (2, 4, 'B');
INSERT INTO TAKES (student_id, section_id, grade) VALUES (3, 1, 'B');
INSERT INTO TAKES (student_id, section_id, grade) VALUES  (3, 5, 'A');
INSERT INTO TAKES (student_id, section_id, grade) VALUES (4, 2, 'A');
INSERT INTO TAKES (student_id, section_id, grade) VALUES (4, 6, 'B');
INSERT INTO TAKES (student_id, section_id, grade) VALUES (5, 3, 'A');
INSERT INTO TAKES (student_id, section_id, grade) VALUES  (5, 7, 'A');
INSERT INTO TAKES (student_id, section_id, grade) VALUES (6, 4, 'B');
INSERT INTO TAKES (student_id, section_id, grade) VALUES  (6, 1, 'A');
```

```
INSERT INTO TAKES (student_id, section_id, grade) VALUES (7, 5, 'A');

INSERT INTO TAKES (student_id, section_id, grade) VALUES  (7, 2, 'B');


INSERT INTO TIME_SLOT (time_slot_id, day, start_time, end_time) VALUES ('A', 'M',
TO_DATE('09:00', 'HH24:MI'), TO_DATE('10:30', 'HH24:MI'));

INSERT INTO TIME_SLOT (time_slot_id, day, start_time, end_time) VALUES ('B', 'T',
TO_DATE('10:30', 'HH24:MI'), TO_DATE('12:00', 'HH24:MI'));

INSERT INTO TIME_SLOT (time_slot_id, day, start_time, end_time) VALUES ('C', 'W',
TO_DATE('13:00', 'HH24:MI'), TO_DATE('14:30', 'HH24:MI'));

INSERT INTO TIME_SLOT (time_slot_id, day, start_time, end_time) VALUES ('D', 'R',
TO_DATE('15:00', 'HH24:MI'), TO_DATE('16:30', 'HH24:MI'));

INSERT INTO TIME_SLOT (time_slot_id, day, start_time, end_time) VALUES ('E', 'F',
TO_DATE('11:00', 'HH24:MI'), TO_DATE('12:30', 'HH24:MI'));

INSERT INTO TIME_SLOT (time_slot_id, day, start_time, end_time) VALUES ('F', 'M',
TO_DATE('14:00', 'HH24:MI'), TO_DATE('15:30', 'HH24:MI'));


INSERT INTO CLASSROOM (building, room_number, capacity) VALUES ('Taylor', '3128', 50);

INSERT INTO CLASSROOM (building, room_number, capacity) VALUES ('Taylor', '3114', 40);

INSERT INTO CLASSROOM (building, room_number, capacity) VALUES ('Watson', '2230', 60);

INSERT INTO CLASSROOM (building, room_number, capacity) VALUES ('Watson', '2245', 45);

INSERT INTO CLASSROOM (building, room_number, capacity) VALUES ('Newton', '1320', 55);

INSERT INTO CLASSROOM (building, room_number, capacity) VALUES ('Newton', '1310', 50);

INSERT INTO CLASSROOM (building, room_number, capacity) VALUES ('Darwin', '2110', 40);
```

iii. Alter the datatypes of any 3 columns

- ⇨ **ALTER TABLE STUDENT MODIFY name VARCHAR2(100);**
- ⇨ **ALTER TABLE INSTRUCTOR MODIFY salary NUMBER(12,2);**
- ⇨ **ALTER TABLE DEPARTMENT MODIFY building VARCHAR2(100);**

iv. Increase the salary of each instructor in the Comp. Sci. department by 10%.

```
=>  UPDATE instructor set salary = salary + salary * .1 where DEPT_ID = 1;
```

v. Delete all courses that have never been offered (that is, do not occur in the section relation).

=> **DELETE FROM COURSE WHERE course_id NOT IN (SELECT DISTINCT course_id FROM SECTION);**

vi. Delete all tuples in the instructor relation for those instructors associated with a department located in the Watson Building.

- **DELETE FROM INSTRUCTOR WHERE dept_id IN (SELECT dept_id FROM DEPARTMENT WHERE building = 'Watson');**

vii. Give a 5% salary raise to those instructors who earn less than 70000.

=> UPDATE instructor set salary = salary + salary * .05 where salary < 80000;

viii. a. Create a new course "CS-001", titled "Weekly Seminar", with 1 credits.

=> insert into course values ('CS-001', 'Weekly Seminar', 1, 1);

b. Create a section of this course in Fall 2017, with sec_id of 1, and with the location of this section not yet specified.

=> insert into section(section_id, course_id, sec_id, semester, year) values (8,'CS-001', 1, 'FALL', 2017);

c. Delete the course CS-001. What will happen if you run this delete statement without first deleting offerings (sections) of this course?

=> delete from course where course_id = 'CS-002';

ix. From a given set of any relational tables, perform the following:

Create Views (with and without check option, with and without read only), Drop views, Selecting from a view.

x. Create any table, use sequence for insertion of the rows and drop the sequence.

⇨ **create table emp(emp_id int primary key, name varchar2(20), mobile int);**

create sequence emp_seq start with 1 increment by 1;

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz1', '9000000001');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz2', '9000000002');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz3', '9000000003');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz4', '9000000004');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz5', '9000000005');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz6', '9000000006');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz7', '9000000007');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz8', '9000000008');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz9', '9000000009');

INSERT INTO emp (emp_id, name, mobile) VALUES (emp_seq.NEXTVAL, 'xyz10','9000000010');

For drop => drop sequence emp_seq;

xi. Rename the above table.

⇨ **rename emp to employee;**
⇨ **alter table employee rename column mobile to mobilenumber;**

xii. Add a new column PINCODE with not null constraints to the above existing table.

⇨ **ALTER TABLE employee ADD pincode INT DEFAULT 0 NOT NULL;**

xiii. Drop the column with the use cascade constraints.

⇨ **ALTER TABLE employee DROP COLUMN pincode CASCADE CONSTRAINTS;**